

CLINICAL AND POPULATION SCIENCES

Polygenic Risk Identifies Older Adults Who May Benefit From Aspirin for the Primary Prevention of Ischemic Stroke

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BACKGROUND: Low-dose aspirin is no longer recommended for routine primary prevention in older adults due to bleeding risks outweighing vascular benefits. We hypothesized that an integrative polygenic score (iPGS) could identify a subgroup of older individuals who derive net benefit from aspirin for the primary prevention of ischemic stroke.

METHODS: We performed post hoc analysis of the ASPREE randomized, placebo-controlled trial (Aspirin in Reducing Events in the Elderly) of daily 100-mg aspirin, in 12 031 genotyped participants of European ancestry aged >70 years without prior cardiovascular disease. The iPGS was derived from >1.2 million variants and evaluated both continuously and by quintiles. Cox models assessed associations between polygenic risk, ischemic stroke, and major bleeding events, and tested the interaction between the iPGS and treatment allocation, with adjustment for baseline lifestyle and clinical covariates.

RESULTS: The mean age of participants was 75.1 years, and 54.9% were women. Over a median of 4.6 years, 187 ischemic strokes and 373 major bleeds occurred, including 101 intracranial bleeds (46 hemorrhagic strokes). Each 1-SD increase in the iPGS was associated with higher incident ischemic stroke risk (hazard ratio, 1.39 [95% CI, 1.20–1.62]). An interaction between the continuous iPGS and aspirin allocation was observed for ischemic stroke ($P=0.04$) but not major bleeding. In the highest iPGS quintile, aspirin reduced ischemic stroke by 51% (hazard ratio, 0.49 [95% CI, 0.28–0.85]) without significantly increasing major bleeding (hazard ratio, 1.15 [95% CI, 0.71–1.88]). No benefit was observed in the overall cohort or in lower-risk quintiles.

CONCLUSIONS: Among older adults, high polygenic risk identifies individuals who may experience substantial stroke reduction with aspirin, with no excess bleeding. These findings raise the possibility that genomic risk stratification may enable targeted aspirin use for the primary prevention of ischemic stroke.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT01038583.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: aspirin ■ genotype ■ ischemic stroke ■ polygenic score ■ randomized controlled trial

Low-dose aspirin has long been an effective therapy for the secondary prevention of ischemic stroke.^{1,2} In the primary prevention setting, however, contemporary guidelines recommend against routine aspirin use because any potential cardiovascular benefits are

offset by increased bleeding risks, particularly intracranial hemorrhage.^{3,4} Recent large-scale trials, including the ASPREE trial (Aspirin in Reducing Events in the Elderly), have reported no overall cardiovascular benefit of aspirin in the primary prevention setting and

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.125.054979>.

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Stroke is available at www.ahajournals.org/journal/str

Nonstandard Abbreviations and Acronyms	
ASPREE	Aspirin in Reducing Events in the Elderly
CAD	coronary artery disease
GWAS	genome-wide association study
HDL-C	high-density lipoprotein-cholesterol
HR	hazard ratio
iPGS	integrative polygenic score
LDL-C	low-density lipoprotein-cholesterol
Lp(a)	lipoprotein(a)
PGS	polygenic score

consistently reported higher rates of major bleeding.^{5–7} A secondary ASPREE analysis focusing on cerebrovascular outcomes further demonstrated a 38% higher incidence of intracranial bleeding among participants randomized to aspirin, without a corresponding reduction in ischemic stroke.⁸ Together, these findings underscore the limitations of generalized aspirin use in older adults. Nevertheless, uncertainty remains as to whether specific subgroups, particularly those at higher underlying risk, may still derive net benefit from aspirin in the primary prevention setting.⁹

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Polygenic scores (PGSs) provide a powerful new tool for identifying higher risk subgroups based on an elevated inherited susceptibility to disease,^{10–12} rather than environmental or age-related risk factors. Our previous work has identified specific genetic subgroups, based on lipoprotein(a) genotypes and PGSs for coronary artery disease (CAD) events, who may derive more benefit than harm from aspirin in primary prevention.^{13–15} Extending this concept to the primary prevention of ischemic stroke, we now investigate the ability of an integrative PGS (iPGS) for ischemic stroke derived from the GIGASTROKE consortium¹² to modify aspirin effects. By integrating randomized trial data with genomic risk stratification, this study aimed to determine whether a precision prevention approach for ischemic stroke could help refine the therapeutic role of aspirin in the primary prevention setting.

METHODS

Data Availability Statement

Data used in this study can be accessed upon request through the ASPREE Principal Investigators. Application details are provided on the ASPREE website (www.ASPREE.org). The data will be made available to investigators whose research proposals have been approved by a designated review committee.

Study Population

The ASPREE trial was a randomized, double-blind, placebo-controlled study conducted in Australia and the United States between 2010 and 2017.^{5,16,17} The trial evaluated whether daily low-dose aspirin (100 mg) prolonged the primary end point of disability-free survival among community-dwelling older adults. A total of 19 114 participants aged ≥70 years (≥65 years for US minorities) were enrolled. Eligible participants had no prior history of diagnosed atherothrombotic cardiovascular disease events, dementia, physical disability, or other known illnesses likely to limit survival to less than 5 years, as verified through clinical records and confirmation by general practitioners. Participants were excluded if they had uncontrolled hypertension, defined as systolic blood pressure ≥180 mmHg or diastolic blood pressure ≥105 mmHg. All participants provided written informed consent. The study was approved by the Human Research Ethics Committees at Monash University and the Alfred Hospital in Australia and by institutional review boards at each US site.

The present study is a post hoc analysis of 12 031 genotyped participants from the ASPREE trial who contributed biospecimens to the ASPREE Healthy Ageing Biobank for genetic analysis (Figure 1). To minimize potential for population stratification bias in the PGS distribution, only individuals of European ancestry were included, as determined by genetic principal component analysis, and overlap with the European reference population from the 1000 Genomes Project (phase 3).¹⁸ Participants of non-European ancestry (<5% of all genotyped participants) were excluded from the study due to insufficient power for ancestry-specific PGS analyses. Individuals aged <70 years at enrollment and close biological relatives (coefficient of relatedness >0.05) were also excluded, as described previously.¹⁵

End Points

The primary end point of this analysis was incident ischemic stroke occurring during the randomized trial period of ASPREE. Ischemic stroke was defined according to World Health Organization criteria as an acute, rapid-onset focal neurological deficit of presumed vascular origin lasting at least 24 hours or resulting in death.^{5,8,19} All events were confirmed by neuroimaging (computed tomography or magnetic resonance imaging) or by autopsy when available. Based on clinical and imaging findings, ischemic strokes were classified into subtypes of large-vessel, small-vessel, cardioembolic, or undetermined origin. Strokes of undetermined cause included those with multiple possible etiologies or incomplete diagnostic evaluation.

In parallel, major bleeding events were evaluated to assess the balance of benefit and harm. Major bleeding included hemorrhagic stroke, symptomatic intracranial bleeding (intracerebral, subdural, or subarachnoid), and clinically significant extracranial bleeding.⁵ Intracranial hemorrhages were centrally adjudicated and included both traumatic and nontraumatic events, provided that they met prespecified clinically significant bleeding criteria. Clinically significant extracranial bleeding was defined as bleeding resulting in transfusion, hospitalization or prolongation of hospitalization, surgery, or death. Accordingly, major bleeding was defined as fatal bleeding, bleeding occurring in a critical site, or bleeding leading to hospitalization or transfusion.

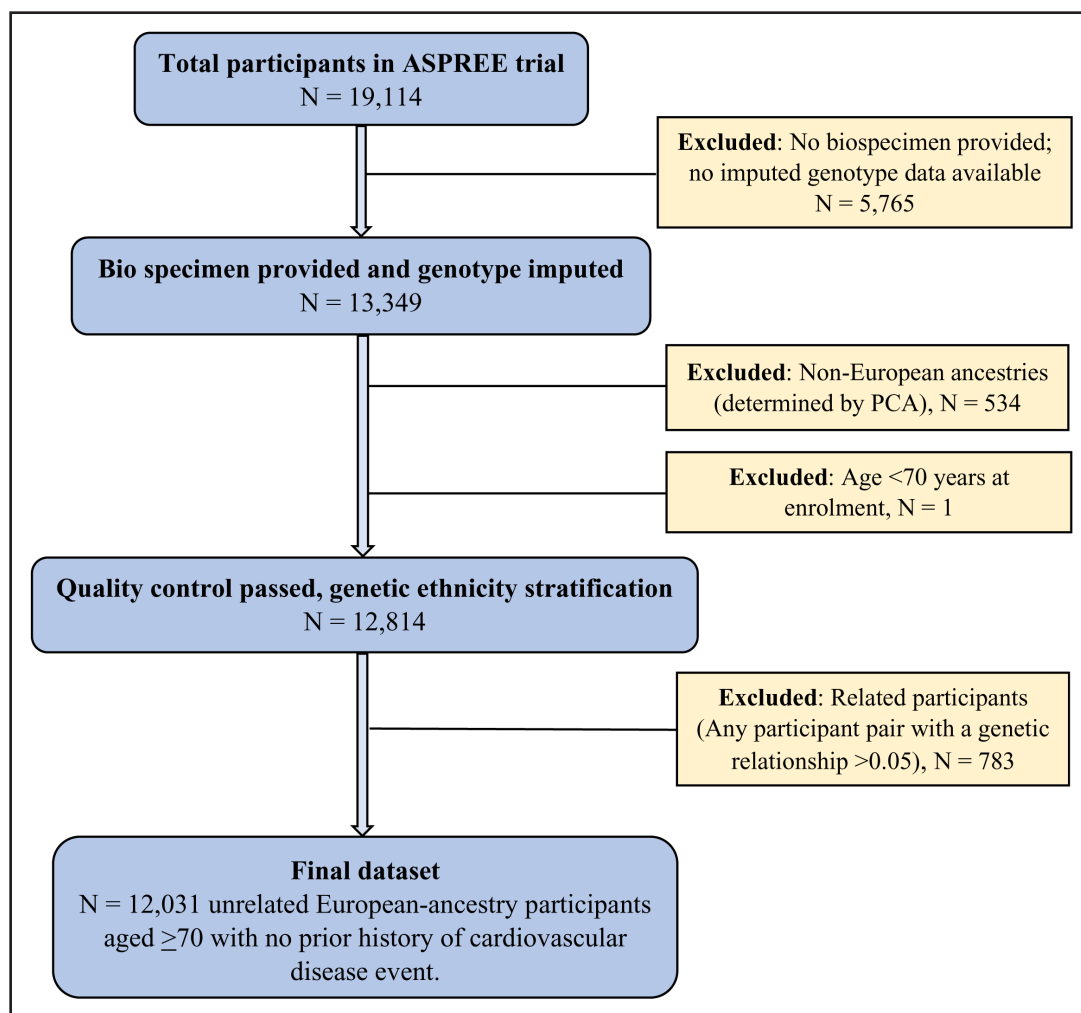


Figure 1. The study flowchart indicates the selection of the final data set.

ASPREE indicates Aspirin in Reducing Events in the Elderly, and PCA, principal component analysis.

All potential events were reviewed and centrally adjudicated by ASPREE Clinical Endpoints Committees, whose members were blinded to participant identity and allocation to aspirin or placebo. Event ascertainment and adjudication procedures were those used in the main ASPREE trial and subsequent cerebrovascular analyses.^{5,8}

Genotyping and PGS

DNA was extracted from peripheral blood or saliva samples collected at baseline and stored in the ASPREE Healthy Ageing Biobank. Genotyping was performed using the Axiom 2.0 Precision Medicine Diversity Array (Thermo Fisher Scientific), which includes ≈800 000 markers optimized for genome-wide coverage and multi-ancestry representation. Genotype calling and quality control were conducted according to standard protocols, as described previously.^{15,19} Variant calling was performed using a custom pipeline aligned to the human reference genome (GRCh38), and imputation was performed based on a diverse reference panel using the TOPMed Imputation Server. Variants with an imputation quality score (R^2) <0.3 were excluded.

For this study, we applied the recently developed and validated iPGS for ischemic stroke, which combines 22

component PGSs derived from genome-wide association studies (GWASs).¹² These included 5 European-specific stroke GWASs, 5 cross-ancestry stroke GWASs, and 12 GWASs of stroke-related traits (atrial fibrillation, coronary artery disease, type 2 diabetes, systolic and diastolic blood pressures, total cholesterol, LDL-C [low-density lipoprotein-cholesterol], HDL-C [high-density lipoprotein-cholesterol], triglycerides, body mass index, height, and smoking behavior). The 12 stroke-related traits and their supporting evidence^{20–27} are summarized in Table S1. Variant weights for the iPGS were obtained from the PGS Catalog²⁸ (ID: PGS002724), comprising 1 213 574 variants. After postimputation quality control in the ASPREE data set, 1 204 449 variants remained and were used to calculate individual iPGS values using PLINK software.²⁹

Statistical Analysis

Baseline characteristics were summarized overall and by quintiles of the ischemic stroke PGS (iPGS). Continuous variables are presented as means with SDs and categorical variables as counts with percentages. Differences across quintiles were evaluated using 1-way ANOVA for continuous variables and χ^2 tests for categorical variables. Event rates were calculated per

1000 person-years of follow-up within each iPGS quintile and treatment group.

Associations between the iPGS and study end points (hazard ratio [HR] and 95% CI) were assessed using Cox proportional-hazards regression models. The iPGS was standardized to a mean of 0 and an SD of 1 and was analyzed both continuously (per SD increase) and categorically by quintiles (Q1–Q5). Q1 served as the reference group. Two nested models were fitted: model 1 adjusted for age at enrollment, sex, and the top 10 genetic principal components, and model 2 further adjusted for baseline lifestyle and clinical covariates, including smoking status, alcohol consumption, body mass index, systolic and diastolic blood pressures, HDL-C, non-HDL-C, hemoglobin, serum creatinine, diabetes, antihypertensive and statin use, and family history of stroke (father, mother, or siblings).

The primary objective was to examine whether polygenic risk modified the effect of aspirin. An interaction term between randomized allocation (aspirin versus placebo) and the iPGS was included in models for ischemic stroke and major bleeding. For categorical analyses, interaction contrasts comparing higher iPGS quintiles with the lowest quintile (Q2–Q5 versus

Q1) were tested. The interaction analyses were adjusted for baseline lifestyle and clinical covariates as specified in model 2. Net clinical benefit was calculated in a simplified manner by presenting the difference between ischemic stroke events averted and major bleeding events induced per 1000 person-years across quintile groups. No formal statistical test was applied to interrogate this metric, and statistical inference for our study was primarily based on the interaction terms from the Cox regression models. All analyses were restricted to the randomized treatment phase. Analyses were performed using R (version 4.3.2). The study followed the STROBE guidelines (Strengthening the Reporting of Observational Studies in Epidemiology; [Supplemental Material](#)).

RESULTS

The baseline characteristics of the 12031 randomized, genotyped participants of the ASPREE trial are shown in Table 1. Their mean (SD) age at baseline was 75.1 (4.2) years, and 6606 (54.9%) were women. A small

Table 1. Baseline Characteristics of ASPREE Participants Included in This Study, Stratified by the iPGS Risk Groups

Baseline characteristics	Overall (N=12031)	Quintile 1 (n=2407)	Quintile 2 (n=2406)	Quintile 3 (n=2406)	Quintile 4 (n=2406)	Quintile 5 (n=2406)	P value*
Age, y; mean (SD)	75.1 (4.2)	75.4 (4.4)	75.2 (4.3)	75.0 (4.2)	74.9 (4.1)	74.8 (4.0)	<0.001
Sex							0.001
Male, n (%)	5425 (45.1%)	1153 (47.9%)	1108 (46.1%)	1093 (45.4%)	1052 (43.7%)	1019 (42.4%)	
Female, n (%)	6606 (54.9%)	1254 (52.1%)	1298 (53.9%)	1313 (54.6%)	1354 (56.3%)	1387 (57.6%)	
Smoking status							0.602
Current, n (%)	367 (3.1%)	71 (2.9%)	74 (3.1%)	62 (2.6%)	81 (3.4%)	79 (3.3%)	
Former, n (%)	4990 (41.5%)	985 (40.9%)	1020 (42.4%)	1009 (41.9%)	1010 (42.0)	966 (40.1%)	
Never, n (%)	6674 (55.5%)	1351 (56.1%)	1312 (54.5%)	1335 (55.5%)	1315 (54.7)	1361 (56.6%)	
Alcohol consumption status							0.488
Current, n (%)	9608 (79.9%)	1939 (80.6%)	1932 (80.3%)	1898 (78.9%)	1941 (80.7%)	1898 (78.9%)	
Former, n (%)	584 (4.9%)	122 (5.1%)	115 (4.8%)	127 (5.3%)	108 (4.5%)	112 (4.7%)	
Never, n (%)	1839 (15.3%)	346 (14.4%)	359 (14.9%)	381 (15.8%)	357 (14.8%)	396 (16.5%)	
HDL-C, mmol/L; mean (SD)	1.6 (0.5)	1.6 (0.5)	1.6 (0.5)	1.6 (0.5)	1.6 (0.5)	1.6 (0.5)	0.004
Non-HDL-C; mmol/L, mean (SD)	3.7 (0.9)	3.7 (0.9)	3.7 (0.9)	3.7 (0.9)	3.7 (0.9)	3.6 (0.9)	0.001
SBP, mmHg; mean (SD)	139.5 (16.3)	137.4 (16.3)	138.7 (15.9)	139.2 (16.4)	140.6 (16.3)	141.6 (16.2)	<0.001
DBP, mmHg; mean (SD)	77.2 (9.9)	77.1 (10.0)	76.9 (9.9)	77.2 (9.8)	77.2 (9.9)	77.6 (10.1)	0.238
Diabetes, n (%)	1108 (9.2%)	147 (6.1%)	171 (7.1%)	218 (9.1%)	269 (11.2%)	303 (12.6%)	<0.001
BMI, kg/m ² ; mean (SD)	28.0 (4.5)	27.8 (4.5)	27.7 (4.4)	28.0 (4.6)	28.1 (4.5)	28.3 (4.7)	<0.001
Hemoglobin, g/dL; mean (SD)	14.2 (1.2)	14.2 (1.2)	14.2 (1.2)	14.2 (1.2)	14.2 (1.2)	14.2 (1.2)	0.914
Creatinine, mg/dL; mean (SD)	0.9 (0.2)	0.9 (0.2)	0.9 (0.2)	0.9 (0.2)	0.9 (0.2)	0.9 (0.2)	0.269
Intake of antihypertensives, n (%)	6097 (50.7%)	927 (38.5%)	1091 (45.3%)	1205 (50.1%)	1373 (57.1%)	1501 (62.4%)	<0.001
Intake of statins, n (%)	3643 (30.3%)	606 (25.2%)	695 (28.9%)	721 (30.0%)	782 (32.5%)	839 (34.9%)	<0.001
Family history of stroke,† n (%)	3579 (29.7%)	649 (27.0%)	672 (27.9%)	734 (30.5%)	769 (32.0%)	755 (31.4%)	<0.001
Treatment allocation							0.330
Aspirin, n (%)	5974 (49.7%)	1207 (50.1%)	1225 (50.9%)	1208 (50.2%)	1167 (48.5%)	1167 (48.5%)	
Placebo, n (%)	6057 (50.3%)	1200 (49.9%)	1181 (49.1%)	1198 (49.8%)	1239 (51.5%)	1239 (51.5%)	

ASPREE indicates Aspirin in Reducing Events in the Elderly; BMI, body mass index; DBP, diastolic blood pressure; HDL-C, high-density lipoprotein-cholesterol; iPGS, integrative polygenic score; and SBP, systolic blood pressure.

*P value was derived by a 1-way ANOVA test for continuous variables or a χ^2 test for categorical variables to compare values in the 5 risk groups.

†Family history includes a history of stroke in the father, mother, or siblings.

proportion were current smokers (3.1%), and 41.5% were former smokers. The iPGS followed an approximately normal distribution (Kolmogorov-Smirnov test; $P=0.95$; Figure S1). Baseline characteristics of the genotyped subset of participants were broadly similar to those of the overall ASPREE cohort and participants not included in the present analysis (Table S2). No major differences were observed in age, sex distribution, or major cardiovascular risk factors.

After stratification into quintiles of the iPGS distribution, several baseline characteristics differed significantly across groups ($P<0.001$), including age, systolic blood pressure, diabetes, body mass index, antihypertensive and statin use, and family history of stroke (Table 1). For example, 6.1% of participants in the lowest iPGS quintile (Q1) had diabetes at baseline compared with 12.6% in the highest quintile (Q5), a >2-fold difference.

During a median follow-up of 4.6 (interquartile range, 3.5–5.6) years, 187 ischemic strokes and 373 major bleeding events occurred among the 12031 participants, including 101 intracranial bleeds of which 46 were hemorrhagic strokes. In fully adjusted models (Table S3), each 1-SD increase in the iPGS was associated with a significantly higher risk of incident ischemic stroke (HR, 1.39 [95% CI, 1.20–1.62]; $P<0.001$). The association was graded across quintiles, with progressively higher risks observed at increasing levels of genetic predisposition (Q5 versus Q1: HR, 2.62 [95% CI, 1.60–4.28]; $P<0.001$).

The effect of aspirin on ischemic stroke differed according to the iPGS distribution (Table 2). The aspirin*×*iPGS interaction was significant when iPGS

was modeled as a continuous variable ($P_{\text{interaction}}=0.037$) and in categorical quintile analyses (Q5 versus Q1; $P_{\text{interaction}}=0.045$) in the fully adjusted model (model 2), indicating that the association between aspirin effect and ischemic stroke varied across levels of genetic risk. In contrast, there was no evidence of interaction between aspirin and iPGS for major bleeding events (all $P>0.05$).

The effect of aspirin on ischemic stroke and major bleeding differed across genetic risk quintiles (Figure 2). Statistical significance for ischemic stroke was observed only in the highest quintile, where stroke risk was reduced by 51% (Q5: HR, 0.49 [95% CI, 0.28–0.85]; $P=0.011$). Reduced stroke risk on aspirin was not observed in the overall cohort (HR, 0.76 [95% CI, 0.56–1.02]; $P=0.063$) or in any of the lower iPGS quintiles (Q1–Q4; $P>0.05$).

Table 3 summarizes the total number of events and the event incidence rates per 1000 person-years, for ischemic stroke and major bleeding events, stratified by iPGS quintiles and treatment allocation. Across the iPGS quintiles, ischemic stroke rates increased progressively with higher genetic risk in the placebo group (from 1.8 per 1000 person-years in Q1 to 7.6 per 1000 person-years in Q5) but showed a flatter gradient among participants receiving aspirin (2.5 in Q1 to 4.0 in Q5). In contrast, major bleeding events in the aspirin group showed a declining trend across quintiles (from 10.7 in Q1 to 7.0 in Q5), indicating that higher genetic risk was not accompanied by an elevated bleeding risk.

To aid clinical interpretation and estimate the overall benefit versus harm of aspirin per subgroup, we descriptively compared the ratio of ischemic stroke events averted to major bleeding events induced per 1000 person-years across iPGS strata. Specifically, we examined the difference between stroke events prevented (vascular benefit) and bleeding events induced (harm) with aspirin before and after genetic risk stratification (Table 3). Using this simple 1:1 event comparison, in the overall genotyped population, we observed aspirin averting 1.0 ischemic stroke events per 1000 person-years and inducing 2.0 bleeding events (for a net harm of –1.0 events). This increase in bleeding on aspirin in the overall cohort was statistically significant (HR, 1.30 [95% CI, 1.05–1.61]; $P=0.014$; Figure 2). However, in the highest iPGS quintile (Q5) specifically, the benefit:harm ratio shifted, with aspirin averting 3.6 ischemic stroke events per 1000 person-years, and inducing only 1.0 bleeding events (for a net benefit of 2.6 events per 1000 person-years).

In the lowest iPGS quintile group (Q1), surprisingly, we observed slightly more ischemic stroke events occurring on aspirin than placebo (–0.7 strokes averted per 1000 person-years). This was further compounded by a higher rate of major bleeding on aspirin in Q1 (4.5 bleeding events induced on aspirin per 1000 person-years) for an overall net harm of –5.2 events per 1000 person-years

Table 2. Interactions Between Aspirin and iPGS for Incident Ischemic Stroke and Major Bleeding Events During the Randomized Controlled Period

Polygenic score	Aspirin <i>×</i> iPGS for ischemic stroke		Aspirin <i>×</i> iPGS for major bleeding	
	P value (model 1)	P value (model 2)	P value (model 1)	P value (model 2)
iPGS (continuous, per SD)	0.044*	0.037*	0.196	0.245
iPGS (categories)				
Quintile 1	Reference	Reference	Reference	Reference
Quintile 2	0.234	0.296	0.425	0.313
Quintile 3	0.300	0.318	0.509	0.361
Quintile 4	0.324	0.300	0.159	0.254
Quintile 5	0.049*	0.045*	0.247	0.268

Cox proportional hazards regression models were used to estimate the HR for the aspirin*×*iPGS interaction with 95% CI. The interaction term was added to model 1, which adjusted for age at recruitment, sex, and top 10 genetic PCs, and model 2, which further adjusted for baseline smoking status, alcohol consumption status, BMI, SBP, DBP, HDL-C, non-HDL-C, hemoglobin, creatinine, diabetes, intake of antihypertensive and statin drugs, and family history of stroke (not for major bleeding). BMI indicates body mass index; DBP, diastolic blood pressure; HDL-C, high-density lipoprotein-cholesterol; HR, hazard ratio; iPGS, integrative polygenic score; PC, principal component; and SBP, systolic blood pressure.

* $P<0.05$.

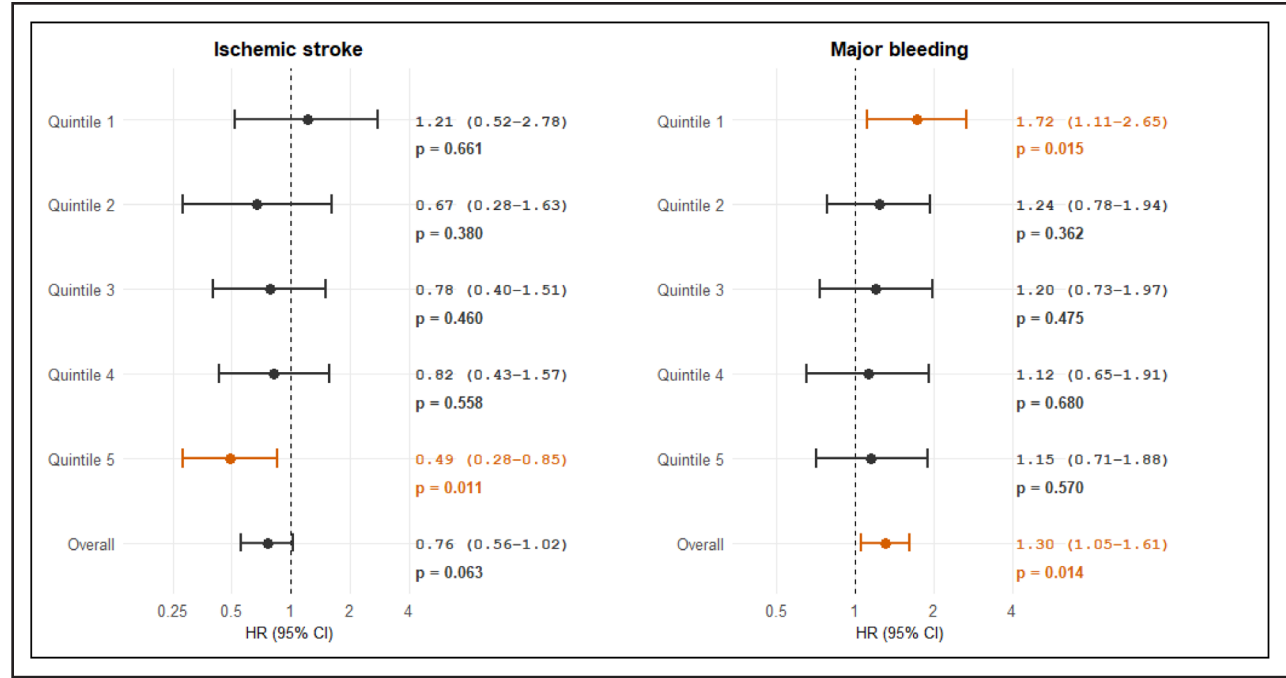


Figure 2. The effect of aspirin on ischemic stroke and major bleeding during the randomized controlled period for subgroups stratified by the integrative polygenic score.

Cox proportional hazards regression models were used to estimate the effect, adjusted for age at recruitment, sex, top 10 genetic principal components, baseline smoking status, alcohol consumption status, body mass index, systolic blood pressure, diastolic blood pressure, HDL-C (high-density lipoprotein-cholesterol), non-HDL-C, hemoglobin, creatinine, diabetes, intake of antihypertensive and statin drugs, and family history of stroke (not for major bleeding). HR indicates hazard ratio.

in Q1. The increase in bleeding events on aspirin in Q1 was statistically significant (HR, 1.72 [95% CI, 1.11–2.65]; $P=0.015$; Figure 2). We further examined the cumulative incidence of events in Q1 compared with Q5, by treatment allocation. As shown in Figure 3, among participants receiving a placebo, those in the highest iPGS quintile (Q5) exhibited a substantially greater cumulative incidence of ischemic stroke compared with those in the lowest quintile (Q1), with divergence in event curves emerging early and widening over time. In contrast,

in the aspirin group, cumulative incidence curves for ischemic stroke between the quintiles remained closely aligned throughout the follow-up period. For major bleeding (Figure S2), no genetic gradient was observed in either treatment group. Notably, within the aspirin group, participants at the highest genetic risk (Q5) had a lower trending cumulative bleeding incidence than those at the lowest risk (Q1), indicating that elevated polygenic risk for ischemic stroke did not correspond to greater bleeding susceptibility in those randomized to aspirin.

Table 3. Ischemic Stroke and Major Bleeding Events During the Randomized Controlled Period, Stratified by Polygenic Risk Quintiles in Aspirin and Placebo Groups Separately

Polygenic risk quintiles	No. of participants		No. of events for ischemic stroke					No. of events for major bleeding					Net benefit:harm ischemic strokes averted-bleeding events induced
	Placebo	Aspirin	Placebo		Aspirin		Ischemic strokes averted per 1000 person-years	Placebo		Aspirin		Bleeding events induced per 1000 person-years	
			Total	Per 1000 person-years	Total	Per 1000 person-years		Total	Per 1000 person-years	Total	Per 1000 person-years		
Quintile 1	1200	1207	10	1.8	14	2.5	−0.7	34	6.2	58	10.7	4.5	−5.2
Quintile 2	1181	1225	13	2.4	9	1.6	0.8	35	6.6	46	8.4	1.8	−1.0
Quintile 3	1198	1208	20	3.7	17	3.1	0.6	30	5.6	42	7.8	2.2	−1.6
Quintile 4	1239	1167	23	4.1	18	3.4	0.7	30	5.4	29	5.5	0.1	0.6
Quintile 5	1239	1167	42	7.6	21	4.0	3.6	33	6.0	36	7.0	1.0	2.6
Overall	6057	5974	108	3.9	79	2.9	1.0	162	5.9	211	7.9	2.0	−1.0

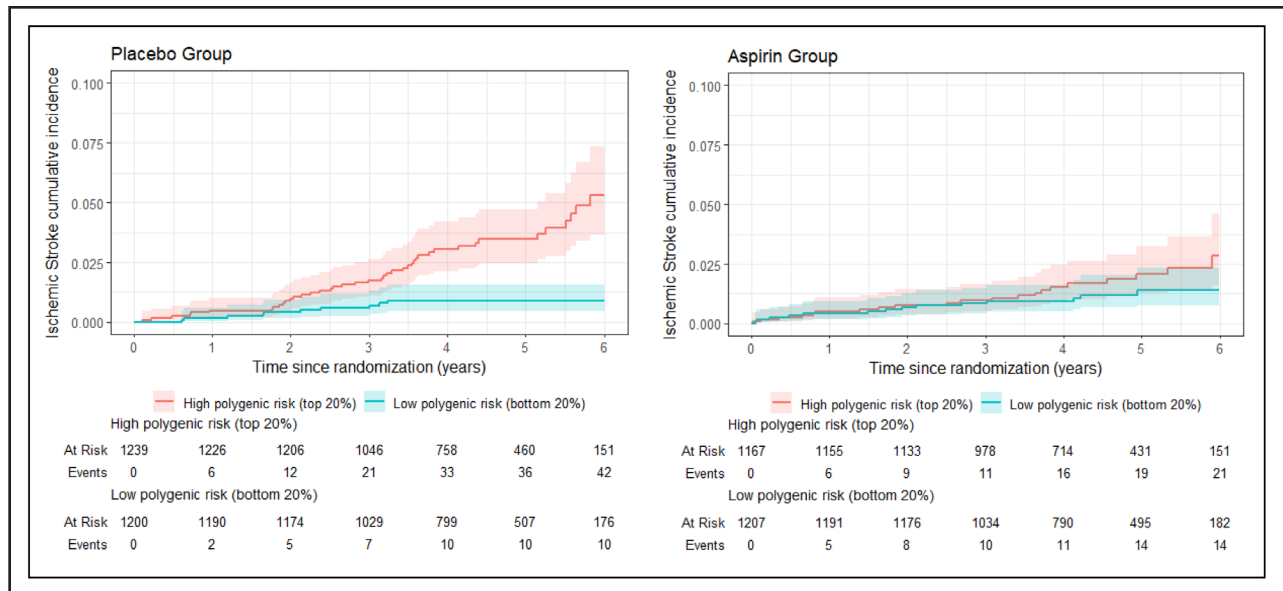


Figure 3. The cumulative incidence of ischemic stroke between participants in the highest (Q5, top 20%) vs lowest (Q1, bottom 20%) quintiles of the integrative polygenic score distribution for placebo and aspirin groups separately.

To determine whether the observed interaction could be explained by previously identified PGS-aspirin signals, we examined the interaction between aspirin allocation and both our published Lp(a) (lipoprotein[a])¹⁴ and CAD¹⁵ PGSs in relation to incident ischemic stroke. Neither score demonstrated a statistically significant interaction with aspirin for ischemic stroke (all $P > 0.05$; Table S4).

DISCUSSION

In this large, randomized trial of older adults without prior cardiovascular disease, we found that a PGS for ischemic stroke modified the effect of aspirin on stroke prevention. Specifically, participants in the highest iPGS quintile experienced a >50% relative reduction in ischemic stroke when randomized to aspirin versus placebo, without evidence of increased major bleeding. This benefit was not observed in the overall trial population or in individuals with lower genetic risk. When the effect was quantified as net benefit versus harm, individuals in the highest-risk iPGS quintile experienced a favorable ratio of 3.6 ischemic strokes averted on aspirin per 1000 person-years, relative to 1.0 major bleeding events induced (for a net benefit of 2.6 events). In the overall cohort and in lower-risk quintiles (Q1–Q3), by contrast, there was net harm. These findings suggest that inherited genetic risk of ischemic stroke may influence the balance of aspirin's vascular benefits versus bleeding harms in older adults, supporting further investigation. Given the post hoc design and lack of formal adjustment for multiple comparisons, our study results should be interpreted with caution and considered to be hypothesis-generating.

Our findings extend a growing evidence base demonstrating that polygenic risk can effectively identify individuals who differentially benefit from preventive therapies. Prior analyses of the ASPREE trial have reported that other genomic risk scores, such as a CAD PGS¹⁵ and Lp(a)-related genotypes,^{13,14} can influence aspirin benefit in the primary prevention setting. Similarly, other studies have reported that statin benefit is greater among individuals with higher genetic risk.¹⁰ By applying a recently developed integrative PGS that captures genetic predisposition across 22 stroke-related traits, the present investigation highlights that inherited susceptibility to ischemic stroke may similarly determine who is most likely to derive vascular protection from aspirin.

Our findings suggest that aspirin's effectiveness and safety profile may meaningfully differ across levels of genetic risk, and that these differences are not apparent (or even detectable) in overall trial analyses without genetic stratification. For example, individuals with low genetic risk of ischemic stroke may derive little or no vascular benefit while remaining exposed to, or even at increased relative risk of, aspirin-related bleeding. In contrast, individuals with higher genetic risk seem to derive significantly greater relative vascular benefits, with relatively lower rates of bleeding for a more favorable benefit-harm profile. These differences were masked in the overall ASPREE trial analysis,⁵ which reported higher rates of bleeding on aspirin across the cohort, and no statistically significant stroke reduction.

A key strength of this analysis is the use of an iPGS that synthesizes genetic information across many pathophysiological domains and risk factors relevant to ischemic stroke, including atrial fibrillation, blood pressure, metabolic traits, lipid fractions, and cardiovascular

disease. It is plausible that individuals with the highest genetic burden are enriched for variants that drive cardioembolic or large-artery stroke phenotypes, which may be more responsive to platelet inhibition. Conversely, individuals with lower genetic risk may harbor fewer biologically active pathways amenable to aspirin's mechanism of action and remain susceptible to its bleeding-related toxicity. Although these mechanistic hypotheses cannot be tested directly in our study, they may provide a framework for understanding why aspirin's effects seem to diverge so sharply across strata of inherited stroke risk.

The observed gradient in benefit also suggests that polygenic information may help address one of the most persistent challenges in primary prevention: identifying individuals in whom the absolute risk of the targeted event is sufficiently high to outweigh treatment-related harms. Traditional clinical risk scores have performed poorly in older adults for this purpose, in part because chronological age dominates risk estimates and obscures heterogeneity in underlying susceptibility. A recent study reported that traditional cardiovascular risk prediction equations, such as the Framingham score and SCORE2-OP (Systematic Coronary Risk Evaluation 2–Older Persons), do not identify individuals in ASPREE who derive net benefit from aspirin.⁹ By contrast, 3 different genetic risk scores have now been shown to identify such subgroups, with a favorable risk/benefit ratio.^{13–15} This implies that genetic risk may capture biological susceptibility that is not reflected in clinical variables or age, allowing for sharper discrimination of individuals who cross the threshold where aspirin's preventive effect outweighs its bleeding risk. From a translational perspective, this raises the possibility that genomic risk stratification may serve as a complementary tool alongside clinical assessment to target aspirin use in select older adults.

Comparison with prior analyses is informative. A previous ASPREE investigation, using an earlier ischemic stroke PGS, found no interaction with aspirin allocation.¹⁹ By contrast, the present study used an updated and more comprehensive iPGS with substantially greater variant representation and integration of cross-ancestry stroke GWAS and related traits, and found an interaction. This difference in PGS construction likely explains the improved ability to detect meaningful gene-treatment interactions in the present study. A similar relative improvement in gene-treatment sensitivity was also observed for earlier and more recent CAD PGSs in ASPREE. The earlier and more conventional CAD PGS was incapable of detecting gene-treatment interactions,³⁰ but a more recent, integrated CAD PGS (also synthesizing information across multiple underlying risk factors) found an interaction with aspirin allocation.¹⁵

Our finding that bleeding risk on aspirin was highest in the lowest genetic risk quintile (Q1) was unexpected. This suggests a subgroup in whom aspirin may be particularly risky. Although the result was not supported by a

statistically significant overall $P_{\text{interaction}}$ value for the continuous iPGS and bleeding events, the Q1 finding raises the possibility that genomic information not only could assist in predicting individuals who benefit from aspirin but also may assist in identifying individuals at disproportionate risk of harm. Bleeding risk is influenced by different clinical factors such as frailty, falls, and medication exposures, which were balanced by randomization but not directly captured by the iPGS. Future research should investigate whether genetic or multiomic profiles related to hemostasis, vascular fragility, or platelet biology may further refine bleeding risk assessment in older adults. We highlight that effective prediction algorithms (genomic or traditional) for major bleeding events are lacking, versus the more established equations for predicting risk for cardiovascular disease events.

Study Limitations

Although different subtypes of ischemic stroke and major bleeding are clinically important^{5,8} and may differ in their genetic determinants or aspirin responsiveness, the number of adjudicated events in ASPREE was too small to support subtype analyses. Once stroke events were divided into categories (eg, disabling/fatal stroke and intracranial versus gastrointestinal bleeding) and further stratified by iPGS quintiles and treatment allocation, group event counts became too low, resulting in insufficient statistical power and unstable estimates. To avoid overinterpretation of underpowered results, we, therefore, restricted our analyses to total ischemic stroke and major bleeding events. Larger studies or pooled data sets may help examine stroke subtype-specific effects. We also acknowledge that our analysis of net benefit versus harm for aspirin, in considering strokes and bleeding events with equal 1:1 weighting, is an oversimplification. While this simple analysis enabled some enumeration of benefits versus harm, it oversimplifies the clinical situation where other factors must be considered.

Another limitation is that the iPGS for ischemic stroke was derived from the GIGASTROKE consortium,¹² which included the ASPREE cohort as one of many contributing data sets. Although ASPREE accounted for <1% of the European ancestry samples, this minor overlap raises a theoretical concern of overfitting. However, this is unlikely to materially affect our findings, as the study's primary aim was to examine gene-treatment interaction in a randomized population (not considered in GIGASTROKE), rather than validate the predictive performance of the iPGS itself. Because the iPGS was used for stratification in randomized comparisons, any minimal overfitting is unlikely to bias the estimated interaction effects.

Additional limitations of the study include its post hoc design and restriction to participants of European ancestry, which may limit generalizability. Future research should assess whether integrating polygenic risk with

traditional risk factors or biomarkers further refines aspirin candidacy and confirm these findings in independent and multiancestry cohorts. While the findings of our study are conceptually promising, we acknowledge that further evaluation and real-world implementation studies would be needed to assess the potential role and clinical utility of using PGSs in primary prevention.

Conclusions

In summary, polygenic risk for ischemic stroke appears to meaningfully modify the balance of vascular benefits versus bleeding harms of aspirin in primary prevention. Individuals with the highest inherited stroke risk experienced substantial benefit on aspirin without increased bleeding, whereas lower-risk individuals accrued net harm. These findings support further investigation of the concept of genomic risk-guided aspirin therapy as a targeted strategy for primary prevention of ischemic stroke. However, the predictive performance of the iPGS needs further validation in external populations, including large non-European ancestry cohorts, to evaluate its generalizability and clinical utility.

ARTICLE INFORMATION

Received December 18, 2025; final revision received April 13, 2026; accepted April 27, 2026.

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Acknowledgments

The authors thank the staff and participants of the ASPREE trial (Aspirin in Reducing Events in the Elderly), and the general practitioners and staff of the medical clinics who cared for the participants.

Sources of Funding

The ASPREE trial (Aspirin in Reducing Events in the Elderly) Biobank is supported by a Flagship Cluster Grant (including the Commonwealth Scientific and Industrial Research Organisation, Monash University, Menzies Research Institute, Australian National University, and The University of Melbourne) and a grant 5U01AG29824-02 from the National Cancer Institute at the National Institutes of Health and Monash University. ASPREE is supported by grants U01AG029824 and U19AG062682 from the National Institute on Aging and the National Cancer Institute at the National Institutes of Health, by grants 334047 and 1127060 from the National Health and Medical Research Council of Australia, and by Monash University and the Victorian Cancer Agency. Dr Yu is supported by a Vanguard Grant (grant 108071-2024_VG) from the National Heart Foundation of Australia. Dr McNeil is supported by a National Health and Medical Research Council of Australia Leadership Fellowship (grant IG1173690). Dr Lacaze is supported by a National Heart Foundation Future Leader Fellowship (grant 107171)

and the National Health and Medical Research Council of Australia Investigator Grant (grant 2026325).

Disclosures

Dr Tonkin reports grants from the National Institutes of Health. Dr Neumann reports grants from Siemens, Abbott Laboratories, PHC, Deutsche Forschungsgemeinschaft; an ownership stake in ART-EMIS Hamburg GmbH; and compensation from Roche Diagnostics GmbH for consultant services. Dr Williamson reports compensation from Alzheimer's Association for consultant services and grants from Alzheimer's Association, the National Institutes of Health, and Biogen, Inc. Dr Eaton reports compensation from Women's Health Initiative for consultant services and compensation from Women's Health Initiative for end point review committee services. Dr Brodtmann reports compensation from Roche, Eli Lilly, and Eisai for consultant services and compensation from Biogen for other services. The other authors report no conflicts.

Supplemental Material

Tables S1–S4
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STROBE Checklist

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